

Market Risk Project Report

ESILV - Market Risk (2025-2026)

Authors

Matthieu HANNA GERGUIS & Renaud DE L'EPINE

Group

A4-IF3

Module Coordinator

Matthieu GARCIN

Professor

Nicolas PESCI

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Repository: https://github.com/MatthieuHannaGerguis/Market_Risk_Project

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Project Overview

This report follows the same flow as our notebook: we go question by question and focus on the essentials: what we did, what we found and how to read the results.

The notebook is meant to stand on its own. It is commented and (we hope) very complete. So if you want to double-check a step, rerun the code or see extra outputs, the notebook should explain everything on its own.

We aimed for the same standard in this report: a clean layout and direct interpretations as well as results while staying consistent with the notebook.

We also used Git and GitHub during the project to keep everything organized and track changes. The repository is here: https://github.com/MatthieuHannaGerguis/Market_Risk_Project

At the start of each question, we quickly give what data we use and if needed, how we prepared it. This way, we know what we are working with without having to guess or scroll back.

We also state our conventions whenever it matters, so there is no confusion. For example, we always say how returns are defined, what we call a “loss”, which confidence level we use and the sign convention (like $\varepsilon \in \{-1, +1\}$ for trade direction).

Question A - Historical VaR (Non-parametric)

In Question A, we will measure the market risk of the Natixis stock using a historical (non-parametric) Value-at-Risk (VaR) approach. We do not assume any parametric distribution for returns: we rely only on what the data tells us.

We use the file `Natixis_stock.txt` which contains daily prices (date and price). We compute the VaR on a one day horizon. In this report we use the tail probability $\alpha = 5\%$ (i.e., 95% confidence) for Question A.

We work with two time windows: (i) 2015-2016 is the insample period (to estimate the VaR), (ii) 2017-2018 is the out of sample period (to validate it in Question A.b).

Convention: We treat a negative return as a loss so the VaR threshold is usually a negative return (left tail).

I. Data preparation and return computation (before Question A.a)

We first load the Natixis price data and compute daily returns. This is basic preprocessing but it is very important. If dates are not read correctly or if prices are not properly converted to numbers, the rest of the work can quickly go wrong.

We will reuse the exact same dataset and the exact same return series later (Questions B and C). Therefore, we clean the data and compute returns once and we keep the same setup for the whole report.

Note: Prices are read from a .txt file meaning the “Value” column is not automatically a float. In the raw file, decimals use a comma thus we replace commas with dots before the conversion.

```
[2]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

#data = pd.read_csv("../data/Natixis_stock.txt", delimiter='\\t',
#header=None, parse_dates=[0]) #we will mainly use main dataframe to analyze data in our whole project
data = pd.read_csv("../data/Natixis_stock.txt", delimiter='\\t',
header=None, parse_dates=[0])
data.columns = ['Date', 'Value'] # rename columns
data['Date'] = pd.to_datetime(data['Date'], format="%d/%m/%Y")

data['Value'] = data['Value'].str.replace(',', '.').astype(float) #retype float number as we read a txt
data = data.sort_values('Date') # Let's sort by date to calculate returns

def Returns(data): #function to go to the next row in the same column(we didn't use pct_change() a pandas functiob)
    data['Returns'] = (data['Value'] - data['Value'].shift(1)) / data['Value'].shift(1)
    return data

data = Returns(data)
print(data)

#the first value of returns is NaN which is logical so we decided to erase the row from our dataset
data = data.dropna()
print(data)

alpha = 0.05 # tail probability / risk level (=>95% confidence level)
value_portfolio = 100000 #value portfolio is the actual value of the portfolio
```

Figure 1: Natixis data preparation and computation of daily returns.

Question A - Historical VaR (Non-parametric)

We loaded the text file and renamed the columns to **Date** and **Value**. We made sure the date column was converted to datetime and the values were read as float to handle comma format issues. We then sorted the dataframe chronologically.

After, we compute simple daily returns with the formula:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

```

      Date  Value  Returns
0  2015-01-02  5.621      NaN
1  2015-01-05  5.424 -0.035047
2  2015-01-06  5.329 -0.017515
3  2015-01-07  5.224 -0.019704
4  2015-01-08  5.453  0.043836
...      ...      ...      ...
1018 2018-12-21  4.045 -0.001481
1019 2018-12-24  4.010 -0.008653
1020 2018-12-27  3.938 -0.017955
1021 2018-12-28  4.088  0.038090
1022 2018-12-31  4.119  0.007583

[1023 rows x 3 columns]
      Date  Value  Returns
1  2015-01-05  5.424 -0.035047
2  2015-01-06  5.329 -0.017515
3  2015-01-07  5.224 -0.019704
4  2015-01-08  5.453  0.043836
5  2015-01-09  5.340 -0.020723
...      ...      ...      ...
1018 2018-12-21  4.045 -0.001481
1019 2018-12-24  4.010 -0.008653
1020 2018-12-27  3.938 -0.017955
1021 2018-12-28  4.088  0.038090
1022 2018-12-31  4.119  0.007583

[1022 rows x 3 columns]
```

Figure 2: Data output (with date, value and computed returns).

For this point, the dataset is ready. We have clean prices, clean dates and one return per trading day. Like we said before we set $\alpha = 0.05$ and a portfolio value (here 100,000) because later we might express risk in monetary terms.

Bonus: As a extra check, we plot the Natixis stock price over time just to see the overall pattern and make sure the series looks consistent.

```
[24]: plt.figure(figsize=(10, 5)) # simple visualization the Natixis stock price over time
plt.plot(data["Date"], data["Value"])
plt.xlabel("Date")
plt.ylabel("Stock price")
plt.title("Natixis stock price evolution")

plt.show()
```

Figure 3: Code used to plot Natixis stock price evolution.

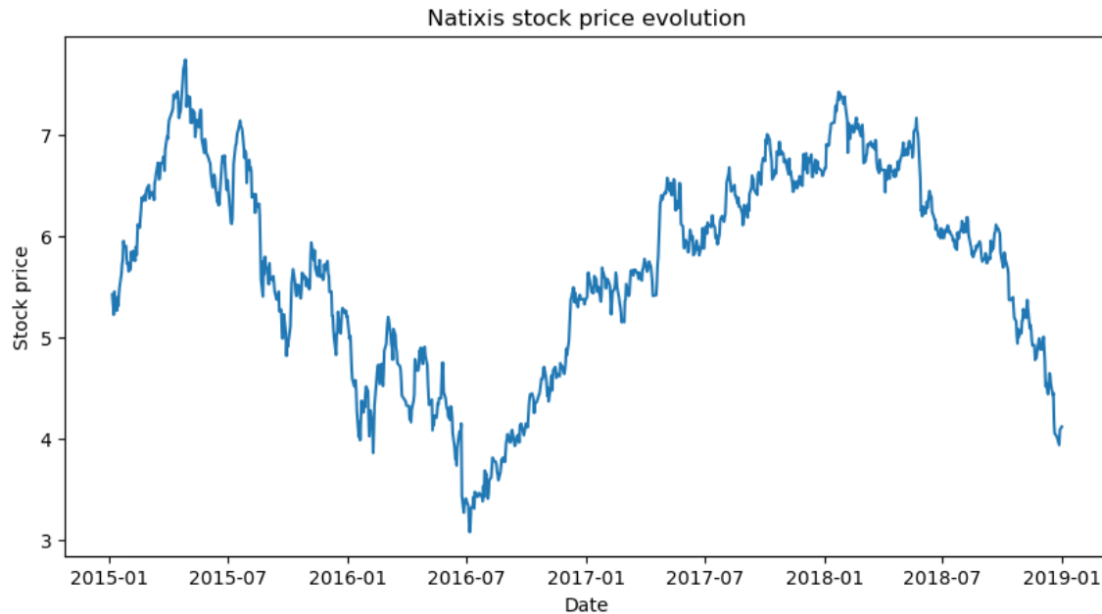


Figure 4: Natixis stock price evolution

Note: To clarify, this plot is a sanity check to help us see that dates are in the right order and that the series looks like a normal price series.

II. Historical VaR estimation (Question A.a)

In this part, we estimate a one-day historical VaR for Natixis returns. We keep the same setup as in Part I: we use daily simple returns, we take $\alpha = 0.05$ (95% confidence), and we use a portfolio value of 100,000 to allow us to express the VaR in euros.

We follow the same logic as in the notebook: first we isolate the in sample period (January 2015 to December 2016) that we use to *estimate* the VaR. Then Question A.b, we use January 2017 to December 2018 to check it out of sample.

```
[25]: # on our sorted dataset we select the time periods required to estimate and validate the historical Var.
data_date_sort_2015 = data[(data["Date"] >= '2015-01-01') & (data["Date"] <= '2016-12-31')]
data_date_sort_2017 = data[(data["Date"] >= '2017-01-01') & (data["Date"] <= '2018-12-31')]

#Once the selection is done now we can sort returns
data_date_sort_2015 = data_date_sort_2015.sort_values("Returns") # using build in sort function of pandas
data_date_sort_2017 = data_date_sort_2017.sort_values("Returns")

print(data_date_sort_2015) #quick check
```

Figure 5: Selecting the periods and sorting returns.

	Date	Value	Returns
378	2016-06-24	3.439	-0.171325
345	2016-05-10	4.083	-0.069083
358	2016-05-27	4.460	-0.061448
405	2016-08-02	3.422	-0.060664
277	2016-02-02	4.208	-0.058613
..	...	...	...
283	2016-02-10	4.101	0.062435
305	2016-03-11	5.083	0.063389
164	2015-08-25	5.755	0.064755
403	2016-07-29	3.685	0.075912
326	2016-04-13	4.785	0.078431

[512 rows x 3 columns]

Figure 6: 2015-2016 returns ranked from worst to best

Course reference: We remember the definition of the historical Value at Risk as introduced during the lectures (slide in Value at Risk, page 68).

Definition

Let X be a random variable depicting a gain. Then, the value at risk with confidence α is:

$$\text{VaR}_\alpha(X) = -F_X^{-1}(1 - \alpha).$$

Alternatively, if Y is a loss ($Y = -X$), $\text{VaR}_\alpha(Y) = F_Y^{-1}(\alpha)$.

Figure 7: Definition of the historical Value at Risk (Course slide page 68).

We now compute the historical VaR at level α . In the notebook, we start with the standard approach based on the empirical quantile. So we simply take the α quantile of the observed returns.

If r is the daily return, the VaR threshold (in return terms) is:

$$\text{VaR}_\alpha^{(\text{ret})} = q_\alpha(r),$$

where $q_\alpha(r)$ is the empirical α -quantile.

Convention: returns can be negative thus $\text{VaR}_\alpha^{(\text{ret})}$ is usually negative (left tail threshold). If we want a positive loss number, we report $-\text{VaR}$ (and in euros we multiply by the portfolio value).

Question A - Historical VaR (Non-parametric)

```
[26]: #historical VaR estimation using the empirical quantile
var_historical = np.quantile(data_date_sort_2015["Returns"], alpha)

var_historical_value = - var_historical * value_portfolio # we have VaR expressed in monetary value
var_historical, var_historical_value

[26]: (np.float64(-0.03777913198405718), np.float64(3777.9131984057176))
```

Figure 8: Computing the empirical VaR (returns and euros).

With $\alpha = 0.05$, the historical VaR on returns is about -0.03778. So the threshold is roughly -3.78% over one day. With a portfolio value of 100,000 this corresponds to a monetary VaR of about 3,777.91 euros.

What does this mean? Over 2015-2016, only 5% of the days have a return below -3.78%. Then, with this historical definition, a "bad" 5% day means losing at least about 3,778 euros.

Moreover, we do the same thing but with a smoother distribution.

Course reference: This is the method showed in the course slides (VaR estimation, slide 69) and it's also what we used in the notebook.

We use the biweight kernel:

$$K(u) = \begin{cases} \frac{15}{16}(1 - u^2)^2 & \text{if } |u| \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

With this kernel the KDE at a point x is:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - r_i}{h}\right),$$

where r_i are the observed returns (2015-2016 here) and h is the bandwidth.

In the code, we set the bandwidth with the "rule of thumb bandwidth estimator"(wikipedia):

A rule-of-thumb bandwidth estimator [\[edit\]](#)

If Gaussian basis functions are used to approximate [univariate](#) data, and the underlying density being estimated is Gaussian, the optimal choice for h (that is, the bandwidth that minimises the [mean integrated squared error](#)) is:^[22]

$$h = \left(\frac{4\hat{\sigma}^5}{3n}\right)^{1/5} \approx 1.06 \hat{\sigma} n^{-1/5},$$

Figure 9: Rule of thumb bandwidth estimator (source: Wikipedia).

where $\hat{\sigma}$ is the sample standard deviation of returns and n is the number of observations


```
[19]: def kernel(u): # biweight pdf
      u = float(u)
      if abs(u) <= 1:
          return (15/16) * (1 - u*u)**2
      return 0

      def kernel_cdf(u): # biweight CDF
          u = float(u)
          if u <= -1:
              return 0
          if u >= 1:
              return 1
          # 0,5 + ∫0^u (15/16)(1-t^2)^2 dt
          return 0.5 + (15/16)*u - (5/8)*u**3 + (3/16)*u**5

      def bandwidth_rule_of_thumb(returns):
          r = np.asarray(returns, dtype=float)
          n = r.size
          sigma = r.std(ddof=1)
          return sigma * (4/(3*n))**0.2 # ≈1.06*sigma*n^(-1/5)

      def kde_cdf(returns, x, h):
          r = np.asarray(returns, dtype=float)
          u = (x - r) / h
          return np.mean([kernel_cdf(ui) for ui in u])

      def VaR_Kernel(data, alpha=0.05):
          r = np.asarray(data["Returns"], dtype=float)
          h = bandwidth_rule_of_thumb(r)
          lo = r.min() - 3*h
          hi = r.max() + 3*h

          for _ in range(60):
              mid = 0.5 * (lo + hi)
              if kde_cdf(r, mid, h) < alpha:
                  lo = mid
              else:
                  hi = mid
          return 0.5 * (lo + hi)

      Value_at_Risk = VaR_Kernel(data_date_sort_2015, 0.05)
      print(f"Value at Risk of Kernel : {Value_at_Risk}")

      Value at Risk of Kernel : -0.03814937022025723
```

Figure 10: KDE based VaR: biweight kernel and 5% quantile.

Question A - Historical VaR (Non-parametric)

The kernel-based VaR (return threshold) is:

$$\text{VaR}_{0.05, \text{kernel}}^{(\text{ret})} \approx -0.03815,$$

so roughly -3.81% over one day.

In euros (same portfolio value 100,000) this corresponds to a loss of about:

$$-\text{VaR}_{0.05, \text{kernel}}^{(\text{ret})} \times 100,000 \approx 3,814.93 \text{ euros.}$$

That is positive by rule.

Finally, the notebook plots the KDE together with the return histogram and the VaR cutoff (vertical line). This is a quick visual check to see where the 5% left tail begins.

```
[51]: plt.figure(figsize=(10, 5)) #we only use this plot to better understand the results obtained above

plt.hist(data_date_sort_2015["Returns"], bins=40, density=True, alpha=0.5, label="Returns")

x_values = np.linspace(
    data_date_sort_2015["Returns"].min(),
    data_date_sort_2015["Returns"].max(),
    500
)

kernel_values = [kernel_density(data_date_sort_2015, x) for x in x_values] # kernel density evaluated on the grid

plt.plot(x_values, kernel_values, label="Kernel density")
#vertical line corresponding to the estimated VaR
plt.axvline(Value_at_Risk, linestyle="--", label="Kernel-based VaR")

plt.xlabel("Returns")
plt.ylabel("Density")
plt.title("Kernel-based Value at Risk")
plt.legend()
plt.show()
```

Figure 11: Code used to plot the biweight kernel density estimate and the kernel-based VaR.

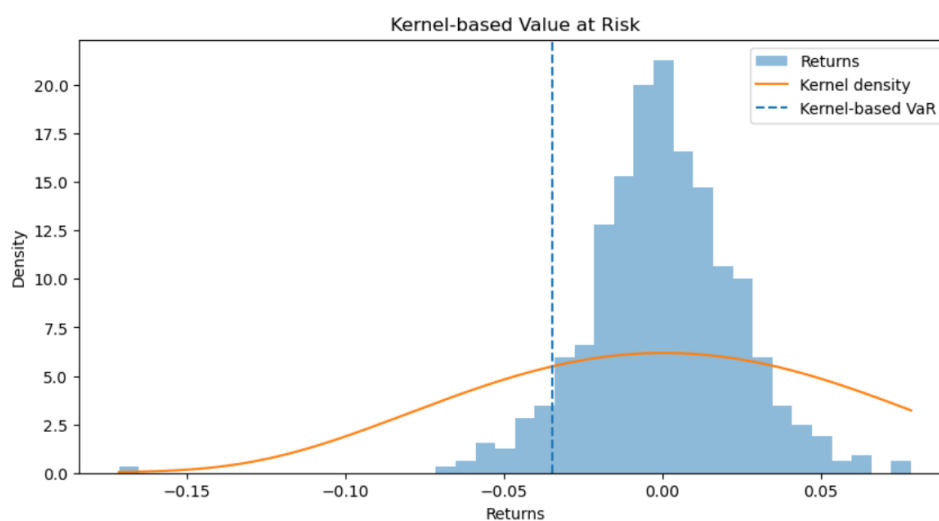


Figure 12: Return KDE and the 5% VaR threshold (2015-2016).

Remark: Here, the kernel VaR is slightly less extreme than the empirical VaR. This can happen because the KDE smooths the distribution thus the left tail is less "step-like".

III. Out-of-sample validation (2017-2018) (Question A.b)

The next step is to check if the non parametric VaR estimated on 2015-2016 behaves well on new data. We take the out of sample period 2017-2018 and count how often returns fall below the VaR threshold.

Conv: A "exceedance" means a return below the VaR threshold because the VaR on returns is a left-tail cutoff. Therefore we count days where $r_t < \text{VaR}_\alpha$.

In the notebook, we do this in a simple loop: we extract the 2017-2018 returns, count how many are below `Value_at_Risk` (the KDE VaR from Part II) and compute the proportion.

```
[10]: #we now look at how many returns exceed the VaR threshold on the 2017-2018 period

data_2017_2018 = data[
    (data['Date'] >= '2017-01-01') &
    (data['Date'] <= '2018-12-31') # out-of-sample period (2017-2018)
]
returns_2017_2018 = data_2017_2018['Returns']

exceedances = 0

for r in returns_2017_2018:
    if r < Value_at_Risk:
        exceedances += 1 # count how many times returns fall below the estimated VaR
exceedance_ratio = exceedances / len(returns_2017_2018) #proportion of exceedances

print("With non-parametric VaR")
print(exceedances, "returns out of", len(returns_2017_2018), "exceed the VaR threshold ")
print("This corresponds to", round(exceedance_ratio * 100, 2), "% for a risk level of 5%")

With non-parametric VaR
8 returns out of 510 exceed the VaR threshold
This corresponds to 1.57 % for a risk level of 5%
```

Figure 13: Counting exceedances in 2017-2018 (KDE VaR)

Looking at the results, we get 8 exceedances out of 510 returns(exceedance rate of 1.57%.) which is below the 5% nominal level

Therefore, the kernel-based VaR is conservative over 2017-2018.

Note: This kind of historical or kernel VaR assumes that past return behavior (2015-2016 here) is still a good proxy for what comes next (2017-2018). In practice, this can fail (volatility changes) which is why out of sample check matters.

Question B - Expected Shortfall

In question B we computed the Expected Shortfall associated with the non-parametric VaR estimated Question A.

We reuse the same return series as in Question A and the same non-parametric VaR threshold we obtained (`Value_at_Risk`).

The idea is simple. VaR gives a cutoff (a quantile of the return distribution). Expected Shortfall goes further by averaging returns that fall below the VaR threshold.

Sign convention: Here we work with returns (not losses). So the left-tail VaR and the ES are negative numbers. In practice, we select the tail with $R \leq VaR_\alpha$ and then ES is the average of these tail returns thus it's also negative.

Course reference: In the lectures, Expected Shortfall is also presented as a spectral (distortion) risk measure. For a continuous distribution, it can be written as:

$$ES_\alpha = \mathbb{E}[R \mid R \leq VaR_\alpha].$$

This is the definition we use here (same as in the notebook).

Expected shortfall

Let X be the gain of the portfolio, $\alpha \in [0, 1]$ a confidence level. We define several risk measures depicting the average of the $1 - \alpha$ worst losses:

- the **worst conditional expectation**: $WCE_\alpha(X) = \sup\{\mathbb{E}[-X|A], \mathbb{P}(A) \geq 1 - \alpha\}$,
- the **tail conditional expectation**: $TCE_\alpha = \mathbb{E}[-X|X \leq -VaR_\alpha(X)]$,
- the **average VaR (AVaR)**, or **expected shortfall (ES)**, or **conditional VaR**:
 $ES_\alpha(X) = \frac{1}{1-\alpha} \int_\alpha^1 VaR_r(X) dr = \frac{1}{1-\alpha} \int_\alpha^1 F_{-X}^{-1}(r) dr.$

Figure 14: Expected Shortfall definition (Course slide page 75).

Computation of the Expected Shortfall:

In practice, we take all in sample returns (2015-2016) that are below (or equal to) the non parametric VaR, and we compute their mean. This is exactly what the notebook does.

```
[20]: # weselect the returns that are below (or equal to) the VaR threshold (worst tail)
      rdt_over_VaR = data_date_sort_2015["Returns"][data_date_sort_2015["Returns"] <= Value_at_Risk]
      #we compute ES empirically:we average the returns in the Left tail (R <= VaR_alpha)
      # which matches ES_alpha = E[R|R <= VaR_alpha].
      ES = np.mean(rdt_over_VaR)

      print("Non parametric VaR:", Value_at_Risk)
      print("Expected Shortfall (non-parametric):", round(ES, 5))

      Non parametric VaR: -0.03814937022025723
      Expected Shortfall (non-parametric): -0.05336
```

Figure 15: code tail selection and ES computation.

Question B - Expected Shortfall

From the notebook output, the non-parametric VaR (95%) is about -3.81% and the ES (95%) is about -5.34%. Indeed, ES averages the returns inside the worst 5% tail, so it is more extreme than VaR in magnitude ($|ES| > |VaR|$).

Question C - Extreme Value Theory (EVT)

In this question, we focus on the analysis of extreme returns using tools from Extreme Value Theory (EVT). We use the Natixis return series and study extreme gains and extreme losses separately to get a clearer view of tail behavior.

We use the same return series as in Question A but now we zoom in on the extreme tails with an EVT approach.

Sign convention: in this section we work in loss terms. When returns are negatif, we define losses = $-$ returns, so losses are positive numbers (EVT VaR is a positive loss)

I. Data preparation and tail separation (EVT setup, before Question C.a)

To apply Extreme Value Theory, we will first split returns into two tails: one for extreme gains and one for extreme losses.

In the notebook, we take positive returns as **gains** and we convert negative returns into positive numbers for **losses**. Next we sort the two series because EVT estimators use order statistics (sorted samples).

```
[31]: gains = data['Returns'][data['Returns'] > 0]
      losses = -data['Returns'][data['Returns'] < 0] #converts losses into positive values thus EVT applies to the right tail.

      gains_sorted = np.sort(gains)
      losses_sorted = np.sort(losses)
```

Figure 16: Tail separation for EVT: gains ($R > 0$) and losses ($L = -R > 0$), then sorts the two

Note: EVT-based VaR naturally comes as a positive loss threshold.

II. Tail index estimation with the Pickands estimator (Question C.a)

In part a, we estimate the extreme value index (shape parameter) ξ for the gain tail and for the loss tail. We use the Pickands estimator which estimates ξ directly. We do not estimate the full GEV location and scale parameters as we do not need them for Question C.b.

Course reference: The notebook follows the formula from the EVT-based VaR estimation slides.

then, the **Pickands estimator**, defined by:

$$\xi_{k(n),n}^P = \frac{1}{\log(2)} \log \left(\frac{X_{n-k(n)+1:n} - X_{n-2k(n)+1:n}}{X_{n-2k(n)+1:n} - X_{n-4k(n)+1:n}} \right)$$

Figure 17: Pickands estimator for the tail index ξ (course slide page 180).

Course reference: The sign of ξ tells us what type of tail we have. As recalled in the EVT lectures Market Risk

Remark

Then:

- *if $\xi > 0$, the GEV is of Fréchet kind,*
- *if $\xi = 0$, the GEV is of Gumbel kind,*
- *if $\xi < 0$, the GEV is of Weibull kind.*

Figure 18: Sign of ξ (Fréchet / Gumbel / Weibull) (course slide page 166).

We apply the Pickands estimator separately to extreme gains and to extreme losses by using the sorted samples built in Part I: `gains_sorted` and `losses_sorted`.

In the notebook, we write a small function `pickands_estimator(extreme_values)`. We set $k = \lfloor \log(n) \rfloor$ (rule of thumb from the TD) to choose how many tail points we use. Then we take the needed order statistic and compute $\hat{\xi}$ with the Pickands formula.

```
[32]: def pickands_estimator(extreme_values):

    #Pickands estimator for the tail index, extreme observations sorted in increasing order.
    n = len(extreme_values)

    # we set k = int(log(n)) (rule of thumb from the Lectures) to choose how many tail points we use.
    k = int(np.log(n))
    x1 = extreme_values[n - k] #required order statistics
    x2 = extreme_values[n - 2 * k]
    x3 = extreme_values[n - 4 * k]

    # we seek the Pickands estimator according to the theoretical formul
    xi_hat = (1 / np.log(2)) * np.log((x1 - x2) / (x2 - x3))
    return xi_hat

    # we estimate the tail index separately for gains and for losses
    xi_gains = pickands_estimator(gains_sorted)
    xi_losses = pickands_estimator(losses_sorted)
    xi_gains, xi_losses

[32]: (np.float64(0.5772338569463286), np.float64(-0.508971577934174))
```

Figure 19: Pickands estimator: ξ for gains and losses (code and output).

From the notebook, we get a positive tail index for extreme gains ($\xi \approx 0.58$) and a negative tail index for extreme losses ($\xi \approx -0.51$). So the two tails do not behave the same way that points to an asymmetric pattern in Natixis returns.

Using the lecture notation: $\xi > 0$ for gains suggests heavy-tailed behavior (Fréchet extremes), while $\xi < 0$ for losses suggests lighter tails (Weibull extremes).

Notes: In standard EVT applications, we often assume returns are independent and identically distributed. Also, the Pickands estimator uses only a limited number of extreme observations. The estimate can be sensitive to sampling variability and to the choice of the intermediate sequence k .

Bonus: we plot the empirical distributions of extreme gains and extreme losses (tail difference suggested by the Pickands estimate).


```
[33]: plt.figure(figsize=(8, 5))

plt.hist(gains_sorted, bins=40, density=True, alpha=0.7,
         label="Extreme gains", edgecolor="black")

plt.hist(losses_sorted, bins=40, density=True, alpha=0.7,
         label="Extreme losses", edgecolor="black")

plt.xlabel("Return magnitude")
plt.ylabel("Density")
plt.title("Empirical distribution of extreme gains and losses")

plt.legend()
plt.grid(True)
plt.show()
```

Figure 20: Code: histogram comparison of extreme gains and extreme losses.

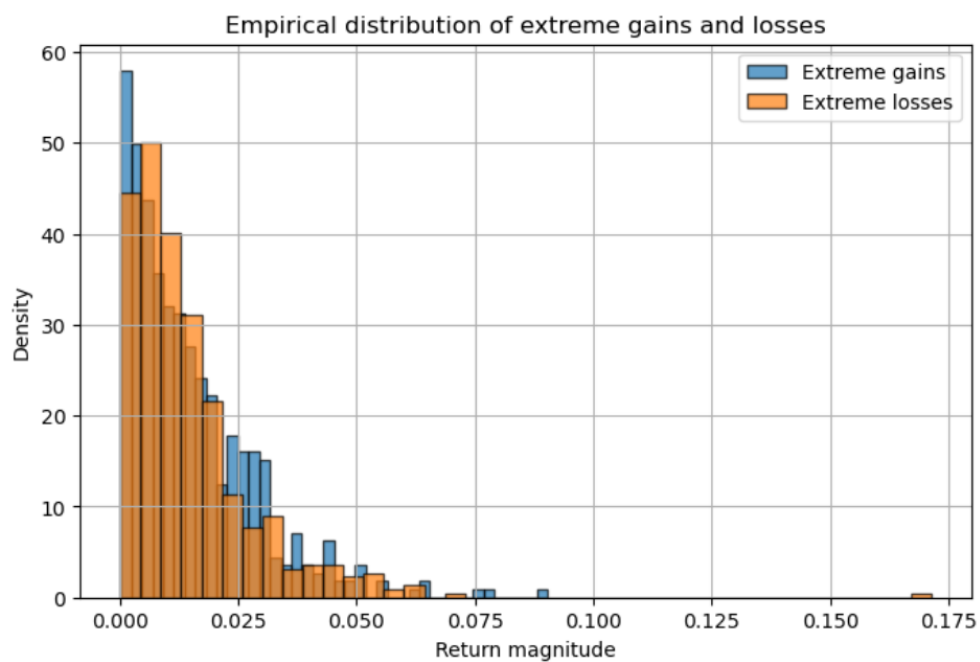


Figure 21: Extreme gains vs extreme losses (empirical distributions).

Comment: the empirical distributions confirm the asymmetrical tail behavior (extreme gains more pronounced)

III. EVT Value at Risk estimation (Question C.b)

In part b, we use Extreme Value Theory to compute Value at Risk estimates from the tail behavior we studied before (Pickands estimator). We focus on the left tail of returns (losses) since VaR is a downside risk measure and we compute the EVT-based VaR using the tail index estimate from Question C.a..

As in standard EVT settings, we assume returns are independent and identically distributed (i.i.d.).

Course ref: From the lectures, VaR is written using the Pickands estimate of the tail index ξ (EVT based VaR estimation slides, page 187).

Pickands estimator

$$VaR(p) = \frac{\left(\frac{k}{n(1-p)}\right)^{\xi^P} - 1}{1 - 2^{-\xi^P}} (X_{n-k+1:n} - X_{n-2k+1:n}) + X_{n-k+1:n},$$

where ξ^P is the Pickands estimator of the GEV parameter.

Figure 22: EVT VaR formula using the Pickands estimator

We now compute the EVT Value at Risk for several confidence levels and compare it to the historical VaR obtained thanks to Question A.

Convention: As mentioned before, we work with losses $L = -R$ (for $R < 0$), so EVT VaR is reported as a positive loss threshold.

In the notebook, we implement a function `var_evt_pickands(extreme_losses, xi_hat, alpha)`. We keep the same choice of $k = \lfloor \log(n) \rfloor$ to stay consistent with Question C.a. Moreover, we will use the needed order statistics and the EVT formula from the course to compute VaR at a given confidence level

Notation: The variable `alpha` in the code is the *confidence level* (for example 0.99), so the tail probability is $1 - \alpha$. That's why the term $(1 - \alpha)$ is in the EVT formula.

```
[34]: def var_evt_pickands(extreme_losses, xi_hat, alpha):

    n = len(extreme_losses)
    #we keep the same choice of k to ensure consistency between the two steps
    k = int(np.log(n))
    # tail order statistics required by the EVT formula
    x_k = extreme_losses[n - k]
    x_2k = extreme_losses[n - 2 * k]

    numerator = (k / (n * (1 - alpha))) ** xi_hat - 1
    denominator = 1 - 2 ** (-xi_hat)
    var_evt = (numerator / denominator) * (x_k - x_2k) + x_k
    return var_evt
```

Figure 23: Code: EVT VaR with Pickands and order statistic.

Subsequently we compute EVT VaR for several confidence levels (90%, 95%, and 99%) and we apply the formula to `losses_sorted` and use the tail index estimate for losses (`xi_losses`).

```
[35]: confidence_levels = [0.90, 0.95, 0.99] # the several confidence levels

for alpha in confidence_levels:
    var_alpha = var_evt_pickands(losses_sorted, xi_losses, alpha)
    print(f"EVT-based VaR (Pickands) at {int(alpha*100)}% confidence level: {var_alpha:.5f}")

EVT-based VaR (Pickands) at 90% confidence level: 0.03141
EVT-based VaR (Pickands) at 95% confidence level: 0.04354
EVT-based VaR (Pickands) at 99% confidence level: 0.05958
```

Figure 24: Pickands EVT VaR: results with 90%, 95%, and 99%.

We can observe that the VaR is going up as we move to higher confidence levels: around 3.1% at 90%, 4.4% at 95%, and about 6.0% at 99%.

Considering that higher confidence levels correspond to deeper tail quantiles this is expected (leads to rarer but more severe losses).

Note: EVT Var uses only the most extreme observations to estimate VaR at high confidence levels. Indeed, we predict losses that are rarer than what we actually see in the data (estimate less reliable).

Compared to the historical VaR from Question A, EVT VaR gives higher importance to the tail of the loss distribution. For that reason, at very high confidence levels it can give higher risk estimates. Additionally, EVT VaR is based on a limited number of extreme observations meaning it can be sensitive to the choice of k and to sampling variability.

Question D - Bouchaud Price Impact Model

We will now estimate the parameters of Bouchaud's price impact model using the TD4 framework.

For this question, we will use the dataset provided for TD4 (`Dataset_TD4.xlsx`) not the Natixis price series used in Questions A to C. The goal of question D is to focus on market impact and liquidity. The inputs are different because we work with intraday variables such as price, volume, bid-ask spread and trade direction (the sign).

Note: in this first part, we keep preprocessing light and we mainly rename columns/do a few quick plots to check the data. We will do a heavier cleaning later when it will be useful for the estimation step (notebook approach).

I. Data preparation (before Question D)

The objective is to obtain a clean, time indexed table containing the key variables required by the model: a price variable, traded volume, bid-ask spread S and the trade sign $\varepsilon \in \{-1, +1\}$.

We start by loading the dataset and inspecting its structure to observe what the raw columns look like and to make sure the file is read correctly.

```
[36]: df_td4 = pd.read_excel("../data/Dataset_TD4.xlsx")
      df_td4.head()
```

	transaction date (1=1day=24 hours)	bid-ask spread	volume of the transaction (if known)	Sign of the transaction	Price (before transaction)
0	0.000202	0.1100	8.0	-1	100.000
1	0.001070	0.1030	NaN	1	99.984
2	0.001496	0.1015	NaN	-1	100.029
3	0.003336	0.0920	NaN	1	99.979
4	0.003952	0.1106	NaN	1	100.060

Figure 25: First rows of `Dataset_TD4.xlsx`.

Then, we create a working dataframe and rename the columns to keep short, clear names that match the model notation and make it easier to read the code and the formulas.

```
[21]: df_vars = df_td4.copy()

df_vars = df_vars.rename(columns={
    "transaction date (1=1day=24 hours)": "time",
    "Price (before transaction)": "price",
    "volume of the transaction (if known)": "volume",
    "bid-ask spread": "spread",
    "Sign of the transaction": "eps" # copy and rename columns without cleaning the dataset
})

df_vars.head()
```

```
[21]:
```

	time	spread	volume	eps	price
0	0.000202	0.1100	8.0	-1	100.000
1	0.001070	0.1030	NaN	1	99.984
2	0.001496	0.1015	NaN	-1	100.029
3	0.003336	0.0920	NaN	1	99.979
4	0.003952	0.1106	NaN	1	100.060

Figure 26: `df_vars` after renaming columns.

Convention: `eps` is the trade sign with the values in $\{-1, +1\}$ (sell vs buy direction)(used in the signed impact model).

Bonus : before estimating the model we created two quick plots that are simple sanity checks to see if the data looks normal.

First, we plot the price evolution over times to helps us see if the price series behaves normally (no obvious data problem).

```
[22]: plt.figure(figsize=(8, 4))
plt.plot(df_vars["time"], df_vars["price"])
plt.xlabel("Time")
plt.ylabel("Price")
plt.title("Price evolution over time")
plt.show()
```

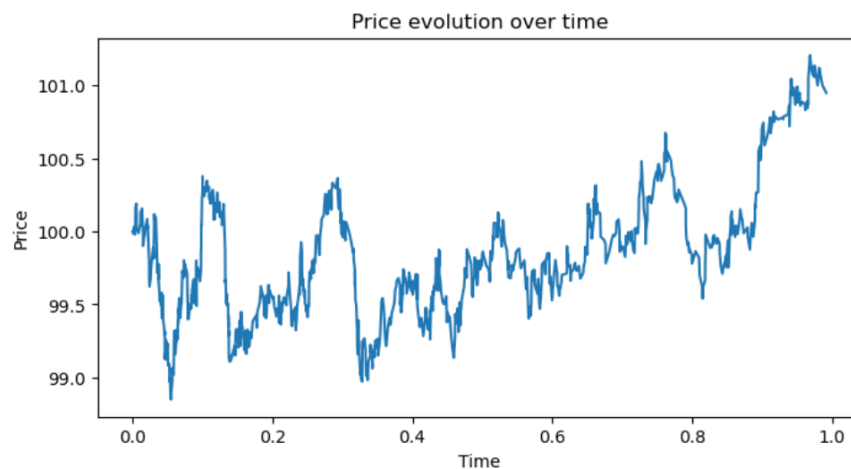


Figure 27: Price evolution over time.

Question D - Bouchaud Price Impact Model

Second, we look at the distribution of trade signs. This allows us to check if buy and sell directions look roughly balanced, an important step before fitting a price impact model.



Figure 28: Distribution of trade signs (buy/sell).

From these plots, we notice that trade signs are fairly balanced notably in line with what we expect before estimating a signed price impact model.

II. Bouchaud price impact model: definition and parameters

In the first part, we introduce Bouchaud's price impact model that is used to estimate how trades move prices.

The Bouchaud models assume today's price is the result of many past trades. Each trade pushes the price up or down (because of the trade sign) and this impact slowly fades away over time. Consequently the model keeps both the direction of trades and how long the impact lasts.

Course reference: we use the transitory impact model shown in the course (page 133 of lecture notes).

Transitory impact model

Backward definition of the price: the price is the sum of past impacts (Bouchaud):

$$p_t = p_{-\infty} + \sum_{s=-\infty}^{t-1} G(t-s) \varepsilon_s S_s V_s',$$

Figure 29: Bouchaud transitory impact model (course slide page 133).

Looking at the slide there's plenty of model parameters. Here is the same simple interpretation as in the notebook:

V : overall impact size. If V is larger, a single trade moves the price more.

r : how impact grows with volume. If $r < 1$, impact is concave (usually observed in data).

γ : how fast impact decays. A larger γ means faster decay (shorter memory). A smaller γ means slower decay (longer memory).

In our implementation, we weight past trades with the impact kernel $G(\ell) = \ell^{-\gamma}$ and a lag cutoff L (avec $L = 50$).

III. Estimation strategy

To estimate the parameters of the transitory impact model the goal is to recover parameters that control: (i) the size of the impact, and (ii) how fast this impact fades away.

We proceed as the following: short term price changes can be explained by the cumulative effect of past signed trades. Thus, we build a predictor from past trades (with a decay over time) and we fit the model by trying to reduce the difference between observed and predicted price changes.

Course reference: the estimation approach follows the TD4 methodology: we choose a parametric form for the impact kernel and we use a regression framework.

1) Dependent variable: price increments

We firstly construct the price increments that will be used as the dependent variable. We define:

$$\Delta p_t = p_t - p_{t-1}.$$

This is what the model tries to explain at a short time scale.

```
[40]: df_est = df_vars.copy()
df_est["delta_price"] = df_est["price"].diff() #we compute price increments from the transaction price
df_est[["time", "price", "delta_price"]].head()
```

	time	price	delta_price
0	0.000202	100.000	NaN
1	0.001070	99.984	-0.016
2	0.001496	100.029	0.045
3	0.003336	99.979	-0.050
4	0.003952	100.060	0.081

Figure 30: Computing Δp_t as first differences of the transaction price.

2) Specification of the impact kernel

To estimate the model we require a parametric form for the impact decay function. Following the method in TD4, we assume a power law decay over the lag. We fix a maximum lag L (here $L = 50$), and we define lags:

$$\ell = 1, 2, \dots, L, \quad G(\ell) = \ell^{-\gamma}.$$

So γ controls the decay speed: larger γ means faster decay.

```
[41]: max_lag = 50
      lags = np.arange(1, max_lag + 1) # lag structure used for the impact kernel

      #parametric pow law form of the impact kernel
      def impact_kernel(lags, gamma):
          return lags ** (-gamma)
```

Figure 31: Power-law impact kernel $G(\ell) = \ell^{-\gamma}$ with max lag $L = 50$.

3) Building the regression signal from past trades

For a given pair (r, γ) , we will build a predictor from past signed trades. We define the signed trade term :

$$u_t = \varepsilon_t S_t Q_t^r,$$

and we aggregate past values with the kernel:

$$X_t = \sum_{\ell=1}^L \ell^{-\gamma} \varepsilon_{t-\ell} S_{t-\ell} Q_{t-\ell}^r.$$

We write traded volume as Q_t (avoid confusion with the impact coefficient V)

On one hand, R controls how impact grows with volume and on the other hand γ controls how impact decays with time.

In the code, we implement past trades with `shift(1)`: shifts the signed term by ℓ steps and add it with weight $G(\ell)$. Missing values created by shifts are replaced by 0 (so we do not use trades that do not exist in the past).

4) Estimating V by OLS for fixed (r, γ)

Once we have X_t and $Y_t = \Delta p_t$ we fit a simple linear relation:

$$Y_t \approx V X_t.$$

For a fixed (r, γ) , the least-squares estimate of V has a closed form:

$$\hat{V} = \frac{\sum_t X_t Y_t}{\sum_t X_t^2}.$$

Then we compute the mean squared error:

$$\text{MSE} = \frac{1}{n} \sum_t (Y_t - \hat{V} X_t)^2.$$

We keep the parameters that give the smallest MSE.

5) Grid search on (r, γ)

We explore a grid of values for r and γ and keep the best fit (smallest MSE). In the code, the grids are:

$$r \in \{0.0, 0.1, \dots, 1.0\}, \quad \gamma \in \{0.1, 0.2, \dots, 2.0\}.$$

Before the grid search we also clean the estimation sample: we keep only rows with non-missing `delta_price`, `eps`, `spread` and `volume` (we also keep basic checks like $\varepsilon \in \{-1, +1\}$, `spread` ≥ 0 , `volume` ≥ 0)

Question D - Bouchaud Price Impact Model

```
[42]: #we only keep trades with known volume here.
df_fit = df_est.copy()
df_fit = df_fit.dropna(subset=["delta_price", "eps", "spread", "volume"]).reset_index(drop=True)

df_fit = df_fit[df_fit["eps"].isin([-1, 1])]
df_fit = df_fit[(df_fit["spread"] >= 0) & (df_fit["volume"] >= 0)]

max_lag = 50 # we keep a reasonable maximum lag for the kernel
lags = np.arange(1, max_lag + 1)

def impact_kernel_power(lags, gamma):
    # impact decay:  $G(L) = L^{-\gamma}$ 
    return lags ** (-gamma)

def build_regressor(df, r, gamma, max_lag=50):
    """
    We build  $X_t$  and we model:  $\text{delta\_price}_t \approx V * X_t$ 
    """
    G = impact_kernel_power(np.arange(1, max_lag + 1), gamma)

    signed_term = df["eps"] * df["spread"] * (df["volume"] ** r)
    X = np.zeros(len(df), dtype=float)
    for l in range(1, max_lag + 1):
        X += G[l - 1] * signed_term.shift(l).fillna(0.0).to_numpy()

    Y = df["delta_price"].to_numpy()
    return X, Y

def estimate_v_ols(X, Y):
    # OLS
    denom = np.dot(X, X)
    if denom == 0:
        return np.nan
    return np.dot(X, Y) / denom

def mse(Y, Y_hat):
    return np.mean((Y - Y_hat) ** 2)

# grid search
r_grid = np.arange(0.0, 1.01, 0.1) # concavity exponent
gamma_grid = np.arange(0.1, 2.01, 0.1) # decay exponent

best = {"mse": np.inf, "V": None, "r": None, "gamma": None} # structure using the best result

for r in r_grid:
    for gamma in gamma_grid:
        X, Y = build_regressor(df_fit, r=r, gamma=gamma, max_lag=max_lag)
        valid = np.isfinite(X) & np.isfinite(Y)
        Xv, Yv = X[valid], Y[valid] # build regression variables

        V_hat = estimate_v_ols(Xv, Yv) # estimate parameters
        if np.isnan(V_hat):
            continue
        Y_hat = V_hat * Xv
        current_mse = mse(Yv, Y_hat)

        if current_mse < best["mse"]: # keep the best parameters
            best = {"mse": current_mse, "V": V_hat, "r": r, "gamma": gamma}

best

[42]: {'mse': np.float64(0.005377107972098853),
      'V': np.float64(0.001992121301477238),
      'r': np.float64(0.7000000000000001),
      'gamma': np.float64(2.0)}
```

Figure 32: Estimation steps: cleaning, kernel based regressor construction, OLS estimate of V , MSE and grid search over (r, γ) .

We report the best fit in a small summary table.

```
[116]: pd.DataFrame([best])
```

```
[116]:
```

	mse	V	r	gamma
0	0.005377	0.001992	0.7	2.0

Figure 33: Best parameters from the grid search (MSE, V , r , γ).

6) Result and interpretation

From the printed results, we get :

$$\text{MSE} \approx 0.005377, \quad \hat{V} \approx 0.001992, \quad \hat{r} = 0.7, \quad \hat{\gamma} = 2.0.$$

Analyzing the screenshot, we remark r is below 1 meaning impact is concave in volume (common empirical feature). In addition γ is high, that is the impact decays fast and is mostly temporary.

This might reflect a liquid market environment, nonetheless it can also point to limits in the parametric form or in the lag window we chose.

Conclusion: To conclude, we obtain estimates for the concavity and the decay of price impact using simple parametric specification of Bouchaud model,. The model captures key features of market impact. The decay estimate lies at the upper bound of the grid ($\gamma = 2.0$) suggesting a boundary solution. A possible interpretation is that the true decay is faster or that the model is too restrictive for this dataset.

Question E - Multiresolution Correlation & Hurst Exponent

In question E we study FX rates with Haar wavelets and the Hurst exponent looking at correlations at different time scales and how volatility scales with time.

Once again, we use a new dataset: `Dataset_TD5.xlsx` that contains high frequency FX prices. Like usual before doing the analysis, we first need to clean the data and build proper time series.

I. Data Description and Preprocessing (Before Question E)

The dataset contains three FX pairs: GBPEUR, SEKEUR and CADEUR.

At each time step, we do not have one single price. Indeed, since we have HIGH and LOW we then build a representative price series for each pair and then compute returns.

To better understand how the file is organized we start by reading the raw Excel file and heading the first lines.

```
[69]: df_td5 = pd.read_excel("../data/Dataset_TD5.xlsx", headers=None)
      df_td5.head(10)
```

```
[69]:
```

	0	1	2	3	4	5	6	7	8	9	10
0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN
1	GBPEUR Curncy	NaN	NaN	NaN	SEKEUR Curncy	NaN	NaN	NaN	CADEUR Curncy	NaN	NaN
2	Date	HIGH	LOW	NaN	Date	HIGH	LOW	NaN	Date	HIGH	LOW
3	2016-03-07 08:59:59.990000	1.2932	1.2917	NaN	2016-03-07 08:59:59.990000	0.10725	0.1072	NaN	2016-03-07 08:59:59.990000	0.6842	0.6829
4	2016-03-07 09:15:00	1.294	1.293	NaN	2016-03-07 09:15:00	0.10728	0.10717	NaN	2016-03-07 09:15:00	0.6849	0.6841
5	2016-03-07 09:30:00	1.2943	1.2922	NaN	2016-03-07 09:30:00	0.10726	0.10719	NaN	2016-03-07 09:30:00	0.6844	0.6837
6	2016-03-07 09:45:00	1.293	1.2913	NaN	2016-03-07 09:45:00	0.10728	0.10721	NaN	2016-03-07 09:45:00	0.6844	0.6839
7	2016-03-07 10:00:00	1.2931	1.2921	NaN	2016-03-07 10:00:00	0.10725	0.10719	NaN	2016-03-07 10:00:00	0.684	0.6835
8	2016-03-07 10:15:00	1.2926	1.2921	NaN	2016-03-07 10:15:00	0.10724	0.10718	NaN	2016-03-07 10:15:00	0.6839	0.6836
9	2016-03-07 10:30:00	1.293	1.2906	NaN	2016-03-07 10:30:00	0.10722	0.1072	NaN	2016-03-07 10:30:00	0.684	0.6834

Figure 34: Raw preview of `Dataset_TD5.xlsx`(three FX pairs are in separate blocks

Looking at the screenshot the file is not a clean table and there are empty or unnamed columns thus we chosen to reorganize everything into one dataframe.

Indeed, we kept only the useful columns and we rename the columns in a clear way: **Date**, **High**, and **Low** for each FX pair.

We also converted the three date columns into a real datetime format crucial to keep correct time ordering.

Question E - Multiresolution Correlation & Hurst Exponent

```
[45]: df_clean = df_td5.iloc[3:, [0, 1, 2, 4, 5, 6, 8, 9, 10]] #we remove the first 3 row of the dataset

df_clean.columns = [
    "Date_GBPEUR", "High_GBPEUR", "Low_GBPEUR",
    "Date_SEKEUR", "High_SEKEUR", "Low_SEKEUR",
    "Date_CADEUR", "High_CADEUR", "Low_CADEUR"
]

df_clean["Date_GBPEUR"] = pd.to_datetime(df_clean["Date_GBPEUR"]) #we rename the column and convert datetime format
df_clean["Date_SEKEUR"] = pd.to_datetime(df_clean["Date_SEKEUR"])
df_clean["Date_CADEUR"] = pd.to_datetime(df_clean["Date_CADEUR"])

df_clean.shape
df_clean.head()
```

	Date_GBPEUR	High_GBPEUR	Low_GBPEUR	Date_SEKEUR	High_SEKEUR	Low_SEKEUR	Date_CADEUR	High_CADEUR	Low_CADEUR
3	2016-03-07 08:59:59.990	1.2932	1.2917	2016-03-07 08:59:59.990	0.10725	0.1072	2016-03-07 08:59:59.990	0.6842	0.6829
4	2016-03-07 09:15:00.000	1.294	1.293	2016-03-07 09:15:00.000	0.10728	0.10717	2016-03-07 09:15:00.000	0.6849	0.6841
5	2016-03-07 09:30:00.000	1.2943	1.2922	2016-03-07 09:30:00.000	0.10726	0.10719	2016-03-07 09:30:00.000	0.6844	0.6837
6	2016-03-07 09:45:00.000	1.293	1.2913	2016-03-07 09:45:00.000	0.10728	0.10721	2016-03-07 09:45:00.000	0.6844	0.6839
7	2016-03-07 10:00:00.000	1.2931	1.2921	2016-03-07 10:00:00.000	0.10725	0.10719	2016-03-07 10:00:00.000	0.684	0.6835

Figure 35: Cleaned table with new names and datetime conversion.

To build one price per FX pair we use the mid-price (average of HIGH and LOW):

$$P_t = \frac{\text{HIGH}_t + \text{LOW}_t}{2}.$$

The formula gives one price per time step and also helps us reduce microstructure noise compared to using only HIGH or only LOW

Then we compute simple returns from these average prices:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}.$$

Finally, we drop the missing values created by the return computation.

```
[46]: df_clean["Avg_GBPEUR"] = (df_clean["High_GBPEUR"] + df_clean["Low_GBPEUR"]) / 2
df_clean["Avg_SEKEUR"] = (df_clean["High_SEKEUR"] + df_clean["Low_SEKEUR"]) / 2 # we work with average prices to reduce microstructure noise
df_clean["Avg_CADEUR"] = (df_clean["High_CADEUR"] + df_clean["Low_CADEUR"]) / 2

df_clean[["Avg_GBPEUR", "Avg_SEKEUR", "Avg_CADEUR"]] = \
df_clean[["Avg_GBPEUR", "Avg_SEKEUR", "Avg_CADEUR"]].astype(float)

df_clean["Ret_GBPEUR"] = df_clean["Avg_GBPEUR"].pct_change() #we compute returns from average prices
df_clean["Ret_SEKEUR"] = df_clean["Avg_SEKEUR"].pct_change()
df_clean["Ret_CADEUR"] = df_clean["Avg_CADEUR"].pct_change()

df_clean = df_clean.dropna()

print(df_clean)
```

Figure 36: Building mid-prices and computing returns (pct_change())

```

      Date_GBPEUR High_GBPEUR Low_GBPEUR      Date_SEKEUR \
4      2016-03-07 09:15:00      1.294      1.293 2016-03-07 09:15:00
5      2016-03-07 09:30:00      1.2943     1.2922 2016-03-07 09:30:00
6      2016-03-07 09:45:00      1.293     1.2913 2016-03-07 09:45:00
7      2016-03-07 10:00:00     1.2931     1.2921 2016-03-07 10:00:00
8      2016-03-07 10:15:00     1.2926     1.2921 2016-03-07 10:15:00
...
12927 2016-09-07 17:00:00     1.1879     1.1867 2016-09-07 17:00:00
12928 2016-09-07 17:15:00     1.1883     1.1874 2016-09-07 17:15:00
12929 2016-09-07 17:30:00     1.188     1.1874 2016-09-07 17:30:00
12930 2016-09-07 17:45:00     1.1874     1.1866 2016-09-07 17:45:00
12931 2016-09-07 18:00:00     1.187     1.1869 2016-09-07 18:00:00

      High_SEKEUR Low_SEKEUR      Date_CADEUR High_CADEUR Low_CADEUR \
4      0.10728     0.10717 2016-03-07 09:15:00     0.6849     0.6841
5      0.10726     0.10719 2016-03-07 09:30:00     0.6844     0.6837
6      0.10728     0.10721 2016-03-07 09:45:00     0.6844     0.6839
7      0.10725     0.10719 2016-03-07 10:00:00     0.684     0.6835
8      0.10724     0.10718 2016-03-07 10:15:00     0.6839     0.6836
...
12927 0.10536     0.10531 2016-09-07 17:00:00     0.6897     0.6893
12928 0.10537     0.10534 2016-09-07 17:15:00     0.6902     0.6895
12929 0.10538     0.10536 2016-09-07 17:30:00     0.6902     0.6898
12930 0.10537     0.10536 2016-09-07 17:45:00     0.6902     0.6901
12931 0.10537     0.10537 2016-09-07 18:00:00     0.6901     0.6901

      Avg_GBPEUR Avg_SEKEUR Avg_CADEUR Ret_GBPEUR Ret_SEKEUR \
4      1.29350     0.107225     0.68450     0.000812     0.000000e+00
5      1.29325     0.107225     0.68405     -0.000193     -1.110223e-16
6      1.29215     0.107245     0.68415     -0.000851     1.865237e-04
7      1.29260     0.107220     0.68375     0.000348     -2.331111e-04
8      1.29235     0.107210     0.68375     -0.000193     -9.326618e-05
...
12927 1.18730     0.105335     0.68950     -0.000589     1.424231e-04
12928 1.18785     0.105355     0.68985     0.000463     1.898704e-04
12929 1.18770     0.105370     0.69000     -0.000126     1.423758e-04
12930 1.18700     0.105365     0.69015     -0.000589     -4.745184e-05
12931 1.18695     0.105370     0.69010     -0.000042     4.745409e-05

      Ret_CADEUR
4      1.389803e-03
5      -6.574142e-04
6      1.461881e-04
7      -5.846671e-04
8      -1.110223e-16
...
12927 3.627131e-04
12928 5.076142e-04
12929 2.174386e-04
12930 2.173913e-04
12931 -7.244802e-05

[12928 rows x 15 columns]

```

Figure 37: Final cleaned dataset

At this point, preprocessing is done. We can move to the rest of Question E since we have clean FX return series at the smallest available time resolution

II. Multiresolution correlations using Haar wavelets (Question E.a)

In this part, we look at how the three FX rates move together depending on the time scale. Using Haar wavelets, we compute multiresolution correlations between GBPEUR, SEKEUR and CADEUR, and we will discuss if an Epps effect appear.

The idea is to separate very short term moves from long term moves. Then we compute correlations scale by scale (level by level) instead of using only one single correlation.

Course reference: we used the multiresolution framework based on the Haar wavelet transform (see lecture notes page 291).

Multiresolution analysis

The Haar wavelet allows an intuitive understanding of wavelets:

- It is a Daubechies wavelet with a single zero moment, so very rudimentary to approximate regular functions.
- The scaling function is $\phi = \mathbf{1}_{[0,1]}$ and the wavelet:

$$\psi(t) = \begin{cases} -1 & \text{if } 0 \leq t < 1/2 \\ 1 & \text{if } 1/2 \leq t < 1 \\ 0 & \text{otherwise.} \end{cases}$$

- The decomposition on ϕ and ψ of a function $a\mathbf{1}_{[0,1/2]} + b\mathbf{1}_{[1/2,1]}$ allows us to understand multiresolution analysis.
- The approximation of daily returns r_t at the 2^j scale, i.e. $\sum_{k \in \mathbb{Z}} \langle r, \phi_{j,k} \rangle \phi_{j,k}(t)$, is therefore based on returns of horizon 2^j calculated each day, with overlap.
- The detail is easily calculated on a date as a difference in returns. It is zero if the returns follow a trend, non-zero if they oscillate around an average.

Figure 38: Haar wavelet and multiresolution analysis idea (course slide page 291).

As correlations only make sense when comparing the same timestamps, we check that the 3 FX series are aligned before computing .

```
[47]: #before we computing correlations, we quickly check that the three FX series are aligned
# on the same timestamps (row by rows). If not we would need to merge on a common time index.
aligned_gbp_sek = (df_clean["Date_GBPEUR"].values == df_clean["Date_SEKEUR"].values).all()
aligned_gbp_cad = (df_clean["Date_GBPEUR"].values == df_clean["Date_CADEUR"].values).all()

print("GBPEUR vs SEKEUR aligned:", aligned_gbp_sek)
print("GBPEUR vs CADEUR aligned:", aligned_gbp_cad)

GBPEUR vs SEKEUR aligned: True
GBPEUR vs CADEUR aligned: True
```

Figure 39: Check that the three date columns are aligned (same timestamps).

We then take the three return series and drop missing values and apply a Haar multiresolution decomposition. At each level, the function averages pairs of points and extracts local fluctuations. Its then repeats this procedure on the average to describe the signal at increasingly longer time horizons.

For each level, we compute correlations between the three FX pairs using the detail coefficients. To compare those levels, we standardize each level series(mean of 0 ,standard dev of 1) before computing correlations.

Remark: the exact Haar normalization is not important here because we standardized the coefficients at each level before computing correlations.

```
[48]: # we take the clean 15min returns and drop NaN so the three series stay perfect aligned
returns = df_clean[["Ret_GBPEUR", "Ret_SEKEUR", "Ret_CADEUR"]].dropna(how="any")

def haar_wavelet_transform(x, levels=3):

    details_per_level = []
    current = x.copy()

    for _ in range(levels):
        current = current[(len(current) // 2) * 2] # Haar keep only an even number of points for Haar pairs
        approx = (current[::2] + current[1::2]) / 2
        detail = (current[::2] - current[1::2]) / 2 # Haar averaging and differencing
        details_per_level.append(detail)
        current = approx # and we keep going on the smoother part

    return details_per_level

wavelet_results = {} # we compute the Haar coeffs for each FX return series
for col in returns.columns:
    wavelet_results[col] = haar_wavelet_transform(returns[col].values, levels=3)

correlations = {} # we compute correlations between the detail coefficients
n_levels = len(wavelet_results["Ret_GBPEUR"])

for level in range(n_levels):
    gbp = wavelet_results["Ret_GBPEUR"][level]
    sek = wavelet_results["Ret_SEKEUR"][level]
    cad = wavelet_results["Ret_CADEUR"][level]
    # we standardize (mean 0 and standard dev 1) so we can compare levels fairly
    gbp = (gbp - gbp.mean()) / gbp.std()
    sek = (sek - sek.mean()) / sek.std()
    cad = (cad - cad.mean()) / cad.std()

    label = f"Level {level + 1}"
    correlations[label] = {
        "GBPEUR-SEKEUR": float(np.corrcoef(gbp, sek)[0, 1]),
        "GBPEUR-CADEUR": float(np.corrcoef(gbp, cad)[0, 1]),
        "SEKEUR-CADEUR": float(np.corrcoef(sek, cad)[0, 1]),
    }

for lvl, corr in correlations.items():
    print(f"{lvl} correlations:")
    for pair, val in corr.items():
        print(f"    {pair}: {val:.4f}")
```

```
Level 1 correlations:
GBPEUR-SEKEUR: 0.1601
GBPEUR-CADEUR: 0.2182
SEKEUR-CADEUR: 0.1742
Level 2 correlations:
GBPEUR-SEKEUR: 0.1400
GBPEUR-CADEUR: 0.3276
SEKEUR-CADEUR: 0.2537
Level 3 correlations:
GBPEUR-SEKEUR: 0.2997
GBPEUR-CADEUR: 0.2260
SEKEUR-CADEUR: 0.1600
```

Figure 40: Haar multiresolution decomposition and correlations by level

With 15-minute data, Level j roughly corresponds to a horizon of about $(2^j) \times 15$ minutes. Level 1 ≈ 30 minutes, Level 2 ≈ 1 hour, Level 3 ≈ 2 hours meaning higher levels mean a longer horizon.

We get the following correlations:

Level 1: We note that correlations are small overall: about 0.16 for GBPEUR-SEKEUR, about 0.17 for SEKEUR-CADEUR and about 0.22 for GBPEUR-CADEUR. Thus at the shortest time scale the three series do not move together that strongly.

Level 2: dependence becomes more visible for some pairs. GBPEUR-CADEUR increases to about 0.33 and SEKEUR-CADEUR increases to about 0.25 while GBPEUR-SEKEUR stays lower around 0.14.

Level 3: the order changes again. GBPEUR-SEKEUR becomes the more correlated pair

at about 0.30 while GBPEUR-CADEUR goes back to about 0.23 and SEKEUR-CADEUR drops to about 0.16. In conclusion, the pattern is not perfectly monotone across levels. Normal because each level looks at a different time scale not just at a simple aggregation.

A simple way to interpret this is through the usual "Epps effect". At very short horizons, although correlations often look weaker they become easier to see once we move to slightly longer time scales.

Conclusion: overall, correlations look weaker at the very highest frequency and they become clearer when we move to extended time scales. We also learned that the most correlated pair can change across levels(confirms that dependence is scale dependent).

We will move to the Hurst exponent estimation to study the scaling behavior of volatility.

III. Estimation of Hurst exponent and volatility scaling (Question E.b)

In part two, we study the scaling behavior of the three FX return series. We estimate the Hurst exponent for GBPEUR, SEKEUR and CADEUR and we use these Hurst estimates to compute annualized volatilities in a way that does not rely on independence.

Course ref: Hurst estimator (pages 251-252)

An estimate of H for a known Gaussian volatility

We define empirical absolute moments on our sample $[0, T]$ with $1/N$ spacings:

$$M_k = \frac{1}{NT} \sum_{i=1}^{NT} |X(i/N) - X((i-1)/N)|^k. \quad (2)$$

M_k is an estimate of $\mathbb{E}(|X(.) - X(. - 1/N)|^k)$, which value is:

$$\mathbb{E}(|X(.) - X(. - 1/N)|^k) = \frac{2^{k/2} \Gamma(\frac{k+1}{2})}{\Gamma(\frac{1}{2})} \sigma^k N^{-kH}. \quad (3)$$

It thus leads to the following estimate of H :

$$\check{H} = - \frac{\log(\sqrt{\pi} M_k / [2^{k/2} \Gamma(\frac{k+1}{2}) \sigma^k])}{k \log(N)},$$

which converges towards H at the rate $O((\sqrt{\varepsilon_N} \log(N))^{-1})$, when N goes to infinity. Whatever k , it depends on σ .

Figure 41: Estimation of H from absolute moments (Course slide pages 251-252).

From now $k = 2$ (estimator then stems from $\mathbb{E}\{(B_H(t) - B_H(s))^2\} = \sigma^2|t - s|^{2H}$). We combine it with the same kind of statistic M'_2 based on observations in the same window but with halved resolution:

$$M'_2 = \frac{2}{NT} \sum_{i=1}^{NT/2} |X(2i/N) - X(2(i-1)/N)|^2.$$

Then:

$$\hat{H} = \frac{1}{2} \log_2 \left(\frac{M'_2}{M_2} \right). \quad (4)$$

$\hat{H}$ converges almost surely toward H .

Figure 42: Final estimator used here (with $k = 2$) (Course slide pages 251-252).

We implement the estimator with $k = 2$ and compute two empirical moments. One using step size 1 (all consecutive points), and one using step size 2 (every two points) and like in the lecture we combine them to get an estimate of H .

Important detail: the estimator is defined for a price process X_t . We build a price-like series by cumulating returns:

$$X_t = \sum_{i=1}^t r_i.$$

This gives a convenient proxy of a log price path over the sample.

We later compute H for the three FX series.

```
[49]: def Hurst_Exponent(X):
    k = 2 # we use the moment of order 2 (same estimator as in the slides)
    T = len(X)

    # quick safety check (we need enough points)
    if T < 4:
        return np.nan

    M_2 = 0 # we compute M_2 with step size = 1
    for i in range(1, T):
        M_2 += abs(X[i] - X[i - 1])**k
    M_2 /= (T - 1)

    # we compute M_2_prime with step size = 2
    M_2_prime = 0
    for i in range(1, T // 2):
        M_2_prime += abs(X[2 * i] - X[2 * i - 2])**k
    M_2_prime /= (T // 2 - 1)

    # apply H = 0.5 * log2(M_2_prime / M_2)
    return 0.5 * np.log2(M_2_prime / M_2)

# we take the 15 min returns from our cleaned dataset
ret_gbp = returns["Ret_GBPEUR"].values.astype(float)
ret_sek = returns["Ret_SEKEUR"].values.astype(float)
ret_cad = returns["Ret_CADEUR"].values.astype(float)

# we build X_t by cumulating returns (price)
X_GBPEUR = np.cumsum(ret_gbp)
X_SEKEUR = np.cumsum(ret_sek)
X_CADEUR = np.cumsum(ret_cad)

# we compute H for each FX rate
Hurst_GBPEUR = Hurst_Exponent(X_GBPEUR)
Hurst_SEKEUR = Hurst_Exponent(X_SEKEUR)
Hurst_CADEUR = Hurst_Exponent(X_CADEUR)

print("Hurst exponent for GBPEUR:", Hurst_GBPEUR)
print("Hurst exponent for SEKEUR:", Hurst_SEKEUR)
print("Hurst exponent for CADEUR:", Hurst_CADEUR)

Hurst exponent for GBPEUR: 0.686860136564295
Hurst exponent for SEKEUR: 0.6810236267055336
Hurst exponent for CADEUR: 0.6810811380086749
```

Figure 43: Hurst exponent estimation for the three FX series (implementation of the $k = 2$ estimator).

We obtain Hurst exponents all above 0.5: around 0.6869 for GBPEUR, 0.6810 for SEKEUR, and 0.6811 for CADEUR. We can say that the three values are close and suggest persistence

compared to the Brownian benchmark ($H = 0.5$).

Course reminder: for a fractional Brownian motion interpretation: $H = 0.5$ corresponds to standard Brownian motion, $H > 0.5$ mean persistent increments, and $H < 0.5$ mean anti-persistent increments (lecture notes page 210).

Increments of B_H of duration τ are stationary: centred Gaussian of variance $\sigma^2|\tau|^{2H}$.

- if $H = 1/2$, the fBm is identical to a standard Brownian motion,
- if $H > 1/2$, the increments are persistent,
- if $H < 1/2$, the increments are anti-persistent.

Figure 44: Interpretation of H

Notes: we interpret these results with caution. because the sample is finite and microstructure effects can affect scaling . Furthermore, we estimate H on a cumulative sum of returns (a price proxy) not directly on log-prices.

Annualized volatility using Hurst scaling

We now compute annualized volatilities using the Hurst scaling law instead of the usual square root of time rule.

Course reference: the scaling law is:

$$\sigma(\Delta t) \propto (\Delta t)^H,$$

thus when we go from Δt to T , we scale volatility like:

$$\sigma(T) = \sigma(\Delta t) \left(\frac{T}{\Delta t} \right)^H.$$

We use this because it does not assume independent returns (lecture notes, page 210).

Here the base step is 15 minutes. To annualize, we use the factor:

$$N = 252 \times 10 \times 4,$$

because we take 252 trading days per year, about 10 trading hours per day and 4 points per hour (15-minute data).

Remark: N is an approximation (we say we use about 10 trading hours per day). A slightly different choice can change the annualized values a bit but probably not the overall comparison across assets.

Therefore:

$$\sigma_{\text{annual}} = \sigma_{15\text{min}} N^H.$$

Question E - Multiresolution Correlation & Hurst Exponent

We compute $\sigma_{15\text{min}}$ as the standard deviation of 15 minutes returns on the sample and we apply the scaling with the estimated H .

```
[50]: def calculate_annualized_volatility(returns_series, hurst_exponent):
# first we just take the average return on our sample
mean_return = sum(returns_series) / len(returns_series)

# then we compute the volatility at the 15 min step
std_return = np.sqrt(
    sum((x - mean_return) ** 2 for x in returns_series) / (len(returns_series)))

# now we scale it up to one year using the Hurst exponent (so we don't rely on independence)
return std_return * (252 * 10 * 4) ** hurst_exponent # for our dataset, we take 252 trading days per year and about 10 trading hours per day(4 point)

annually_vol_GBPEUR = calculate_annualized_volatility(
    returns["Ret_GBPEUR"].dropna().values.astype(float),
    Hurst_GBPEUR) #we apply the same formule to each FX returns series
print(f"Annualized volatility for GBPEUR: {annually_vol_GBPEUR:.6f}")

annually_vol_SEKEUR = calculate_annualized_volatility(
    returns["Ret_SEKEUR"].dropna().values.astype(float),
    Hurst_SEKEUR)
print(f"Annualized volatility for SEKEUR: {annually_vol_SEKEUR:.6f}")

annually_vol_CADEUR = calculate_annualized_volatility(
    returns["Ret_CADEUR"].dropna().values.astype(float),
    Hurst_CADEUR)
print(f"Annualized volatility for CADEUR: {annually_vol_CADEUR:.6f}")

Annualized volatility for GBPEUR: 0.350331
Annualized volatility for SEKEUR: 0.174243
Annualized volatility for CADEUR: 0.269808
```

Figure 45: Annualized volatility using Hurst scaling (15 minute steps scaled to one year).

We obtain annualized volatilities of about 0.3503 for GBPEUR, 0.1742 for SEKEUR and 0.2698 for CADEUR.

Conclusion (E.b): we find similar Hurst exponents for the three FX series(all above 0.5) This suggests some persistence keeping in mind the limits of intraday data and the estimation method. Using H scaling allows volatility to be anualised beyond the standard square root rule.

Overall, the results from question E show that FX behavior depends on the time scale meaning it should be taken into account when analyzing dependence and risk.

General conclusion (key results)

Question	Main results
A (VaR)	Empirical 1 day VaR ($\alpha = 0.05$): $\text{VaR} \approx -0.037779 \Rightarrow \approx 3,777.91$ euros (portfolio of 100,000).
A(KDE VaR)	Kernel VaR (biweight KDE): $\text{VaR} \approx -0.0381493 \Rightarrow \approx 3,814.93$ euros. Out-of-sample exceedances (2017-2018): $8/510 \approx 1.57\%$.
B (ES)	Expected Shortfall ($\alpha = 0.05$): $\text{ES} \approx -0.05336$.
C (EVT)	Pickands tail index: $\xi_{\text{gains}} \approx 0.5772$ (heavy tail), $\xi_{\text{losses}} \approx -0.05336$ (lighter tail). EVT VaR (losses): $90\% \approx 0.03141$, $95\% \approx 0.04354$, $99\% \approx 0.05958$.
D (Impact)	Bouchaud model estimates: $r \approx 0.7$, $\gamma \approx 2.0$, $V \approx 0.001992$, $\text{MSE} \approx 0.005377$.
E (Wavelets)	Haar multiresolution correlations: weaker at the finest scale, clearer at longer scales (Epps effect).
E (Hurst)	Hurst: $H_{\text{GBPEUR}} \approx 0.6869$, $H_{\text{SEKEUR}} \approx 0.6810$, $H_{\text{CADEUR}} \approx 0.6811$; Annualized vol (H-scaling): $\text{GBPEUR} \approx 0.3503$, $\text{SEKEUR} \approx 0.1742$, $\text{CADEUR} \approx 0.2698$.